
Dr. Sven Berger

Postdoctoral Scientist

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Profile

Postdoctoral scientist at Helmholtz-Zentrum Hereon working at the intersection of materials modelling, corrosion, fluid dynamics, machine learning, research software engineering, and research data management. Active in open-source scientific software and digital research infrastructure, with public project roles in OpenModel, CorWiz, HiRSE_PS, MetaSurf, and MATRIX.

Experience

Postdoctoral Scientist

February 2021 - Present

Helmholtz-Zentrum Hereon, Institute of Surface Science, Department of Interface Modelling

- Research in interface and materials modelling with a focus on corrosion, fluid dynamics, and data-driven workflows.
- Works on machine learning, research software engineering, and research data management.
- Public project roles include OpenModel, CorWiz, HiRSE_PS, MetaSurf (Primary Contact), and MATRIX (PI).
- Contributes to open-source simulation and data tools, including TrixiParticles.jl and CorWiz.

Research Assistant

February 2015 - January 2021

Institute of Aerodynamics, RWTH Aachen University

- Researched computational aeroacoustics, combustion engines, renewable fuels, turbulent mixing, and particle-laden flows.
- Performed software engineering, including performance optimization, benchmarking, testing infrastructure, and code review.
- Helped modernize the code base and development workflow using Git, continuous integration, and a migration to modern C++.
- Completed doctoral research on massively parallel large-eddy simulation for internal combustion engines.

Student Research Assistant

June 2012 - January 2015

Institute of Aerodynamics, RWTH Aachen University

- Developed a discontinuous Galerkin-based computational aeroacoustics solver.

Student Research Assistant

March 2012 - December 2012

Institute of Polymer Processing, RWTH Aachen University

- Worked on finite-element modelling of composite materials and simulation of injection-moulding processes.

Intern (Fluid Engineering)

September 2011 - February 2012

DEUTZ AG

- Developed an OpenFOAM GUI for common simulation tasks and implemented a flash-boiling model.

Community Service

August 2006 - August 2007

Roman Catholic Church

Intern

December 2005 - January 2006

Bayer Technology Services

Education

Doctor of Philosophy (PhD)

2021

RWTH Aachen University

- Thesis: Massively parallel large-eddy simulation solver for internal combustion engines.

Master of Science (M.Sc.), Computational Engineering Science

2007 - 2014

RWTH Aachen University

Selected Projects and Software

- **MATRIX** - PI at Hereon for the ScienceServe project on digital twins and high-performance simulation workflows.
- **MetaSurf** - Primary Contact for the HMC project on interoperable metadata and research-data workflows in surface science (2024-2026).
- **CorWiz** - Hereon project for a corrosion knowledge web service (2023-2024).
- **OpenModel** - Hereon contact for the H2020 project on integrated open-access materials modelling innovation.
- **TrixiParticles.jl** - Open-source Julia framework for particle-based multiphysics simulation; public author and principal developer role documented in the repository and JOSS paper.

Selected Publications

- 1 Neher, N. S.; Faulhaber, E.; Berger, S.; Gassner, G. J.; Schlottke-Lakemper, M. (2025). TrixiParticles.jl: Particle-based multiphysics simulation in Julia. *Journal of Open Source Software*, 10(105), 7044. DOI: 10.21105/joss.07044
- 2 Sohn, S.; Shi, S.; Markmann, J.; Berger, S.; Weissmuller, J. (2025). Compressive behavior and connecting topology of monolithic nanoporous niobium. *Materials Research Letters*, 13(1), 76-85. DOI: 10.1080/21663831.2024.2421867
- 3 Wiese, B.; Berger, S.; Bohlen, J.; Nienaber, M.; Hoche, D. (2023). Property design of extruded magnesium-gadolinium alloys through machine learning. *Materials Today Communications*, 36, 106566. DOI: 10.1016/j.mtcomm.2023.106566
- 4 Berger, S. C.; Wegmann, T.; Meinke, M.; Schroder, W. (2020). Large-Eddy Simulation Study of Biofuel Injection in an Optical Direct Injection Engine. *SAE Technical Paper 2020-01-2121*. DOI: 10.4271/2020-01-2121
- 5 Schlottke-Lakemper, M. A.; Yu, H.; Berger, S. C.; Meinke, M.; Schroder, W. (2017). A fully coupled hybrid computational aeroacoustics method on hierarchical Cartesian meshes. *Computers & Fluids*, 144, 137-153. DOI: 10.1016/j.compfluid.2016.12.001

Skills

Materials Modelling, Corrosion, Interface Modelling, Fluid Dynamics, Computational Aeroacoustics, Machine Learning, Research Software Engineering, Research Data Management, Julia, C++, Python, Git / CI

Languages

German (Native or Bilingual), English (Full Professional), French (Elementary)

Honors and Awards

- RWTH Aachen Dean's List (2014)
- Sun Microsystems Java Star (2005)